Federated Full-Envelope LPV Identification from Incomplete Local Coverage

***Abstract*—Vehicles may supply identification data from narrow speed regions while the fleet collectively covers a wider interval. Pooling can supply missing scheduling information but mix incompatible dynamics. We study federated linear-parameter-varying (LPV) identification under these joint limitations. Clients estimate fixed-speed models by output-error fitting of measured state trajectories and known applied steering, then exchange additive regression statistics and assignment scores to learn shared group maps. Across ten independent nonlinear vehicle fleets with family labels withheld, mean matrix error at locally unobserved speeds is 2.05%, compared with 3.94% for Smooth Local and 4.48% for Global. Both prespecified paired comparisons support an aggregate improvement, although some recipients deteriorate. The distributed regression agrees with a pooled solve for the same assignments. Coverage diagnostics show that accurate local estimates alone do not ensure reliable group extrapolation. Supporting scheduled linear-quadratic-integral (LQI) tests reuse the frozen maps and satisfy the tested constraints in all federated rollouts. Tracking gains over the fixed nominal controller are consistent across fleets; gains over Global are modest and nonuniform. The evidence concerns simulated maneuvers with available states, not observer robustness or a general stability guarantee.**

***Index Terms*—federated identification, LPV systems, incomplete operating coverage, heterogeneous vehicles**

I. Introduction

Model-based vehicle control requires dynamics models that remain useful as operating conditions change. For lateral motion, longitudinal speed affects the relationship between steering, sideslip, and yaw rate. A model identified near one operating point may therefore be inadequate elsewhere. LPV models represent this dependence explicitly, making identification across the intended speed range a central requirement [1], [2].

Obtaining that coverage separately for every vehicle is a demanding calibration task. A vehicle may provide measurements over only a narrow part of its intended operating range. Those measurements constrain the observed response, but a flexible speed-dependent model can remain undetermined outside that region. Repeating measurements at the same speeds reduces noise without resolving missing scheduling directions. The resulting problem is therefore incomplete operating information as well as limited sample count.

Fleet collaboration offers a way to fill these gaps: one vehicle's unobserved speeds may be covered by others. Yet differences in mass, inertia, and tire properties make their measurements only approximately compatible. Pooling all vehicles can introduce bias, while separating them too aggressively can leave each group without sufficient speed coverage. What is needed is selective sharing that supplies missing operating information without overwhelming a recipient's dynamics with incompatible data. Crucially, the partners must be chosen from partial observations, before complete local models are available.

We address this need through federated LPV identification. Vehicles retain calibration records locally, exchange additive statistics, and evaluate candidate group maps at their observed speeds. Alternating assignment and regression learns shared maps over the wider interval. The central question is whether groups inferred from partial data can provide both complementary coverage and a useful approximation of their recipients' dynamics. Figure 1 summarizes the procedure.

Existing work supplies important components of this approach. Local LPV identification constructs scheduling-dependent maps from operating-point models [3]. Federated learning provides a framework for collaboration without centralizing records [4], and FedSysID characterizes the effect of dynamical heterogeneity on collaborative identification [5]. IFCA alternates cluster assignment and model optimization [6]. Clustered SysID applies this structure to LTI systems with identical dynamics within each true cluster [7]. IC-SYSID adds incremental clustering and stability promotion [8], while [9] uses uncertainty-aware grouping for vehicle controller design.

Our distinction is the joint requirement of compatibility and complementary scheduling information. Each client's fixed speed support leaves its LPV regression deficient, and shared maps remain approximate within groups. The cited recovery guarantees do not establish recovery in this setting. We specialize the established alternating clustering structure to this problem and make the following contributions:

1) We formulate LPV identification from incomplete local speed coverage in common physical-state coordinates. A conditional rank statement distinguishes information supplied by pooling from extrapolation imposed by a local prior.
2) We implement label-free group regression using additive statistics and local assignment scores, without requiring complete local LPV maps. Agreement with a pooled regression for the same assignments verifies the distributed implementation.
3) We establish controlled vehicle-fleet evidence for the coverage–bias tradeoff through comparisons with local priors and global pooling, learned-group and recipient audits, and diagnostics of estimation error and group coverage. Supporting LQI tests reuse the frozen OE-derived maps to assess tracking, constraint satisfaction, and recipient tradeoffs under a common controller design.

Sections II and III present the identification problem and federated regression. Section IV describes the benchmark. Sections V and VI report identification and LQI tracking results. Section VII discusses the findings and concludes.

II. Identification Problem

A. Local identification from measured trajectories

Consider M vehicles indexed by $i \in \{1, \dots, M\}$ and the target speed interval $\mathcal{V} = [v_{\min}, v_{\max}]$, with $0 < v_{\min} < v_{\max}$. Vehicle i has state $x_i = [\beta_i, r_i]^\top \in \mathbb{R}^2$, comprising sideslip angle β_i and yaw rate r_i, and front-wheel steering input $u_i \in \mathbb{R}$. Its measurements are $y_{i,k} = x_{i,k} + \nu_{i,k}$, where $\nu_{i,k} \in \mathbb{R}^2$ is state measurement noise and k indexes samples. Applied steering and longitudinal speed are assumed known.

Vehicle i supplies trajectories at the finite speed set $\mathcal{V}_i = \{v_{i,1}, \dots, v_{i,n_i}\} \subset \mathcal{V}$, where n_i is the number of observed speeds. At each speed v, the fitting window contains $N_i(v)$ consecutive transitions with measured states $y_{i,0}, \dots, y_{i,N_i(v)}$ and applied inputs $u_{i,0}, \dots, u_{i,N_i(v)-1}$. Speed is constant within each trajectory. Section IV specifies the maneuvers and fixed nominal acquisition controller.

We identify a two-state affine model by output error (OE), comparing its simulated trajectory with the measurements. For one vehicle–speed pair, suppress i and v and write

$$\widehat{x}_{k+1} = A\widehat{x}_k + Bu_k + c, \qquad \widehat{x}_0 = \zeta, \tag{1}$$

where $A \in \mathbb{R}^{2\times 2}$, $B \in \mathbb{R}^{2\times 1}$, $c \in \mathbb{R}^2$, and $\zeta \in \mathbb{R}^2$ are fitted jointly. Define $W = \operatorname{diag}(\sigma_\beta^{-1}, \sigma_r^{-1})$, where $\sigma_\beta, \sigma_r > 0$ are the specified state measurement noise standard deviations. With $N = N_i(v)$, the fitting objective is

$$\mathcal{L}(A, B, c, \zeta) = \frac{1}{2(N+1)} \sum_{k=0}^{N} \|W(\widehat{x}_k - y_k)\|_2^2, \tag{2}$$

where $\|\cdot\|_2$ is the Euclidean norm and $\widehat{x}_k$ follows (1) without resetting to measured states. Four initializations, including the common nominal model, are optimized; the converged solution with the lowest training loss supplies $\widehat{A}_i(v), \widehat{B}_i(v)$. The offset c avoids requiring a known equilibrium center. Although each maneuver starts at zero lateral state, ζ must describe the fitting window's initial state after settling.

Let $A_i(v) \in \mathbb{R}^{2\times 2}$ and $B_i(v) \in \mathbb{R}^{2\times 1}$ denote the reference local perturbation matrices about the prescribed cornering condition. Collect their entries in $m_i(v) \in \mathbb{R}^{1\times 6}$ in the order $[A_{i,11}, A_{i,12}, A_{i,21}, A_{i,22}, B_{i,1}, B_{i,2}]$. The same ordering of the OE estimates gives $\widetilde{m}_i(v)$. These estimates can differ from $m_i(v)$ through measurement noise, finite-amplitude approximation, and numerical optimization. Only the matrix estimates enter the scheduling dataset $\mathcal{D}_i = \{(v, \widetilde{m}_i(v)) : v \in \mathcal{V}_i\}$; the fitted offsets and initial states remain local.

B. Incomplete scheduling information and shared maps

Let $\phi(v) \in \mathbb{R}^L$ contain L linearly independent scheduling functions shared by all vehicles. Define $\Phi_i \in \mathbb{R}^{n_i \times L}$ and $Y_i \in \mathbb{R}^{n_i \times 6}$ with rows $\phi(v_{i,\ell})^\top$ and $\widetilde{m}_i(v_{i,\ell})$, respectively, for $\ell = 1, \dots, n_i$. The setting of interest has

$$\operatorname{rank} \Phi_i < L. \tag{3}$$

A nonzero $U_i \in \mathbb{R}^{L\times 6}$ can then satisfy $\Phi_i U_i = 0$: coefficient maps differing by U_i agree at observed speeds but can predict differently elsewhere. Additional samples at the same speeds cannot remove this scheduling ambiguity. Local dynamical excitation and scheduling coverage are therefore separate requirements.

For K shared maps, let $g_i \in \{1, \dots, K\}$ assign vehicle i to a group with coefficients $\Theta_{g_i} \in \mathbb{R}^{L\times 6}$. Write

$$m_i(v) = \phi(v)^\top \Theta_{g_i} + d_i(v), \tag{4}$$

where $d_i(v) \in \mathbb{R}^{1\times 6}$ combines basis approximation error and individual departures from shared dynamics. Complementary speed supports can supply missing scheduling directions, but pooling introduces this recipient-dependent mismatch, separate from local estimation error in Y_i.

C. Identification objective and scope

Given $\mathcal{D}_i$, we seek K, assignments g_i, and coefficients Θ_g yielding $\widehat{m}_i(v) = \phi(v)^\top \widehat{\Theta}_{g_i}$. Section III learns these maps from observed targets; evaluation tests accuracy at locally unobserved speeds $\mathcal{V} \setminus \mathcal{V}_i$ and the mismatch introduced by pooling.

Trajectories and local estimates stay on device; clients exchange the statistics and scores defined in Section III. Recipient-specific parameters, reference matrices, and family labels enter neither local OE fitting nor group assignment and regression. Each fresh fleet learns maps for its contributing vehicles. Generalization concerns speeds, not transfer of a fixed library to new recipients. "Full envelope" denotes the speed interval along the prescribed cornering curve, not an arbitrary nonlinear state/input domain.

III. Federated Compatible Regression

The local OE estimates in Section II supply Y_i; Φ_i describes their observed speeds. We alternate shared LPV fitting and vehicle assignment, then select the group count by penalized residual error. OE offsets and initial states remain local; neither unobserved-speed targets nor family labels enter the search.

A. Additive fitting and conditional identifiability

For a fixed partition, let $\mathcal{I}_g = \{i : g_i = g\}$ be the set of vehicles in group g. Its map minimizes the regularized least-squares objective

$$\widehat{\Theta}_g = \arg\min_{\Theta \in \mathbb{R}^{L\times 6}} \left\{ \sum_{i \in \mathcal{I}_g} \|Y_i - \Phi_i \Theta\|_F^2 + \rho \|\Theta\|_F^2 \right\}, \tag{5}$$

where $\|\cdot\|_F$ is the Frobenius norm and $\rho = 10^{-8} + 10^{-12}$ combines ridge regularization with a numerical diagonal adjustment. Each speed–matrix target has equal weight; OE estimation covariances are not used. Client i computes $G_i =$

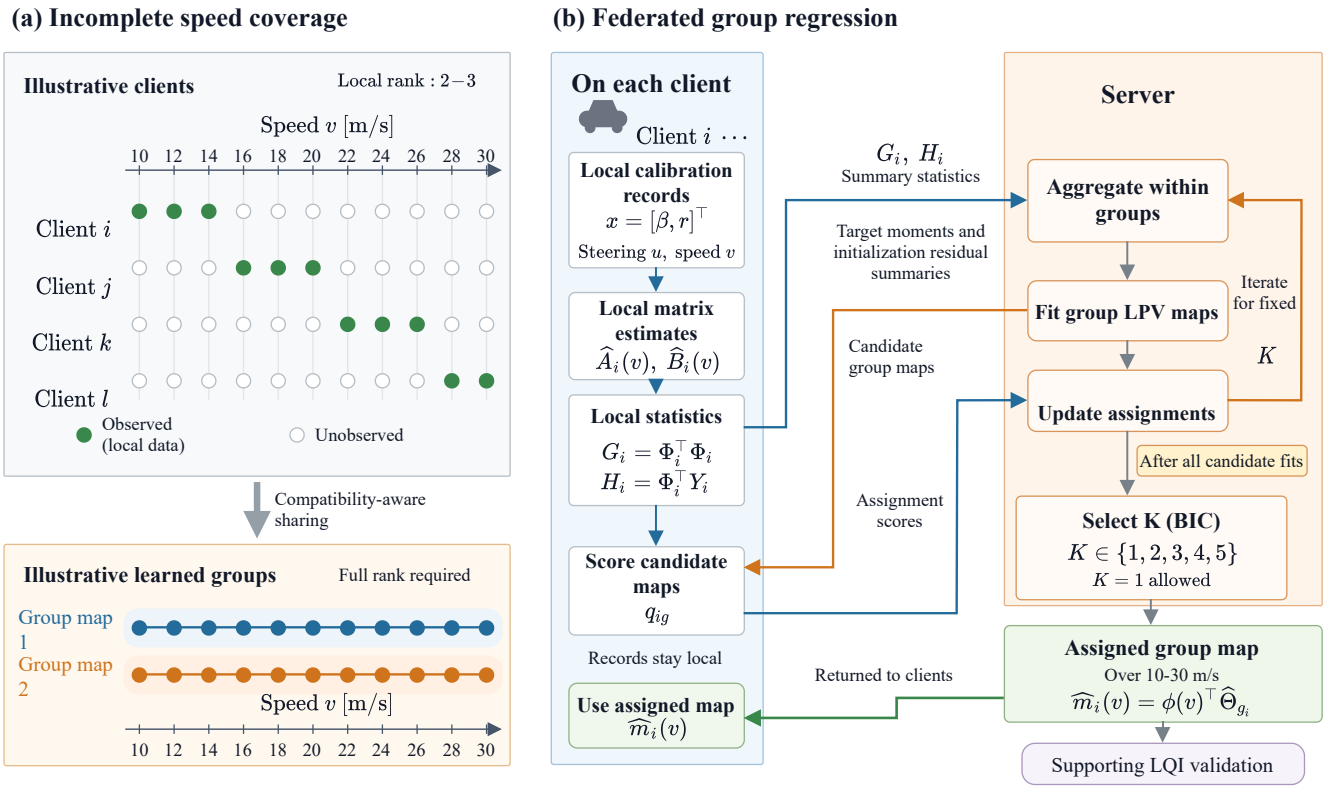


Fig. 1. Compatible federated LPV identification. (a) Illustrative clients and groups. Local speed blocks give scheduling rank two or three for a seven-function basis; sharing can supply missing directions when the pooled group design has rank seven. (b) Clients retain records and matrix targets, exchange additive statistics, residual summaries, and assignment scores, and receive group maps. Fitting and assignment alternate for each candidate K, followed by selection using the Bayesian information criterion (BIC). Local output-error fits use measured state trajectories and known applied steering at fixed speeds. The envelope is 10–30 m/s along the prescribed cornering curve; LQI checks downstream usability.

$\Phi_i^\top \Phi_i \in \mathbb{R}^{L \times L}$ and $H_i = \Phi_i^\top Y_i \in \mathbb{R}^{L \times 6}$. The server sums these statistics within each group and solves

$$\left(\sum_{i \in \mathcal{I}_g} G_i + \rho I_L \right) \widehat{\Theta}_g = \sum_{i \in \mathcal{I}_g} H_i, \tag{6}$$

where I_L is the $L \times L$ identity matrix. For a fixed partition, this gives the pooled ridge solution in exact arithmetic without sharing Y_i. This equivalence concerns regression for the same groups, not agreement between independent clustering runs. Positive regularization ensures a unique solution; identification from data requires a separate rank condition.

Proposition 1: For a fixed nonempty group $\mathcal{I}_g$, let Φ_g be the vertical stack of its local designs Φ_i. The unregularized least-squares fit ($\rho = 0$) has a unique minimizer if and only if $\operatorname{rank} \Phi_g = L$, equivalently $\operatorname{rank}(\sum_{i \in \mathcal{I}_g} G_i) = L$. If, in addition, the targets satisfy $Y_i = \Phi_i \Theta_g$ for every group member, that minimizer equals Θ_g.

Proof: The aggregate Gram matrix is $\Phi_g^\top \Phi_g$, which is positive definite exactly when Φ_g has full column rank. Otherwise, a nonzero $U \in \mathbb{R}^{L \times 6}$ with $\Phi_g U = 0$ can be added to any minimizer without changing its predictions. For exact shared-map targets, Θ_g achieves zero residual and is therefore the unique minimizer in the full-rank case.

Pooling can resolve the missing scheduling directions in Section II. Full rank alone does not control conditioning, OE error, or the mismatch $d_i(v)$. Exact recovery requires unregularized fitting and exact shared-map targets, unlike the regularized fits of noisy targets used here.

B. Assignment and model selection

Assignment balances the six matrix entries by scaling. For entry $j \in \{1, \dots, 6\}$, let τ_j be its standard deviation across all observed target rows, using the population divisor $\sum_i n_i$. The server computes these values from local row counts, entry sums, and squared sums. Define $s_j = \max\{\tau_j, 0.05 \operatorname{median}(\tau_1, \dots, \tau_6)\}$ and $D = \operatorname{diag}(s_1, \dots, s_6)$. Unlike the state-noise weights W in OE, D describes target variation across the fleet. Its entries are checked to be positive and fixed during each search. Given the broadcast maps, client i returns

$$q_{ig} = \|(Y_i - \Phi_i \widehat{\Theta}_g) D^{-1}\|_F^2, \quad g = 1, \dots, K, \tag{7}$$

and is assigned to the lowest-scoring group, with ties choosing the lowest index. Fitting uses the unscaled objective (5); we do not claim that these alternating steps minimize one common objective.

Algorithm 1: Federated Group Learning from Local OE Models

Input: Local designs Φ_i and OE matrix targets Y_i, $i = 1, \dots, M$; common basis and regularization ρ.
Output: Selected K, assignments g_i, and maps $\widehat{\Theta}_g$, $g = 1, \dots, K$.
Prepare statistics and initialization summaries
Clients: send $G_i = \Phi_i^\top \Phi_i$, $H_i = \Phi_i^\top Y_i$, row counts, and target entry sums and squared sums
Server: fit Global $\widehat{\Theta}_0$ by (6) with all vehicles; compute D (Section III-B); broadcast both
Clients: send mean residuals a_i from (8)
Server: standardize the entries of a_i across clients
Search group counts and initial assignments
for $K = 1, \dots, 5$ **do**
  Use one start for $K = 1$ and 12 seeded starts otherwise
  for *each start* **do**
    Server: set $g_i = 1$ if $K = 1$; otherwise initialize by K-means on the standardized summaries
    for $t = 1, \dots, 30$ **do**
      Discard this start if any group has fewer than five vehicles
      Clients: send G_i, H_i and current group tags
      Server: fit each group by (6); broadcast all maps
      Clients: return scores q_{ig} from (7) for every group
      Server: assign each i to the lowest-scoring group, choosing the lowest index in a tie
      Break if no assignment changed
    Discard this start if any final group has fewer than five vehicles
    If the last iteration changed assignments, refit final groups and recompute scores without reassignment
    Server: compute BIC by (9); save this candidate
  Keep the lowest-BIC candidate for this K, if any
Select and distribute the result
Server: select the lowest-BIC candidate across K; send assignments g_i; clients use their selected broadcast maps

For initialization, fit all vehicles as one group to obtain the Global map $\widehat{\Theta}_0$. Client i sends its six-entry mean residual

$$a_i = \frac{1}{n_i} \sum_{\ell=1}^{n_i} \left[\widetilde{m}_i(v_{i,\ell}) - \phi(v_{i,\ell})^\top \widehat{\Theta}_0 \right]. \tag{8}$$

The server standardizes each entry across clients to zero mean and unit variance, uses unit scale for constant entries, and applies K-means. This scaling is separate from D. No complete local LPV map is needed.

Algorithm 1 specifies the search. Define the residual sum $\mathrm{SSE} = \sum_{i=1}^{M} q_{i,g_i}$ and scalar target count $n = 6 \sum_i n_i$. The Bayesian information criterion (BIC) based score is

$$\mathrm{BIC} = n \log\left(\max\left\{\mathrm{SSE}/n, 10^{-12}\right\}\right) + p_K \log n, \tag{9}$$

where $p_K = 6LK + K - 1$ counts map coefficients plus a nominal mixture-weight term. Mixture weights are not estimated, and entries of an OE target need not be independent. Thus, this score is a complexity penalty rather than a likelihood-based guarantee. The lowest score selects the restart and K, including Global pooling at $K = 1$. Neither global optimality nor physical-family recovery is guaranteed. Group rank and conditioning are diagnosed after selection; minimum group size does not ensure full rank, and deficient groups are not replaced by a fallback model.

C. Exchanged information

The messages G_i and H_i contain $L(L+1)/2$ distinct Gram entries and $6L$ right-hand-side entries: 28 and 42 for $L = 7$. Initialization also uses one row count, twelve target moments, and six residual-summary entries per client. Each assignment step uses K client scores and broadcasts K maps of $6L$ entries, with group tags and scales supplied as needed. These payloads do not grow with trajectory length, but all candidate counts and restarts require communication. We make no claim of communication efficiency. Trajectories and matrix targets stay local in the distributed emulator; shared statistics can still reveal information about them. No secure aggregation or differential privacy is implemented.

IV. Benchmark and Evaluation

Local fitting, global pooling, and learned groups use identical OE targets. Acquisition, estimators, and comparisons are fixed after three pilot fleets and tested on ten independent confirmation fleets.

A. Vehicle fleet and local speed coverage

The plant is a single-track model with state $x = [\beta, r]^\top$, where β is the vehicle sideslip angle at the center of mass and r is the yaw rate. The input u is the front-wheel steering angle, and the longitudinal speed v is held constant during each trajectory. The dynamics include nonlinear tire forces:

$$\dot{\beta} = (F_f + F_r)/(mv) - r, \tag{10}$$

$$\dot{r} = (l_f F_f - l_r F_r)/I_z, \tag{11}$$

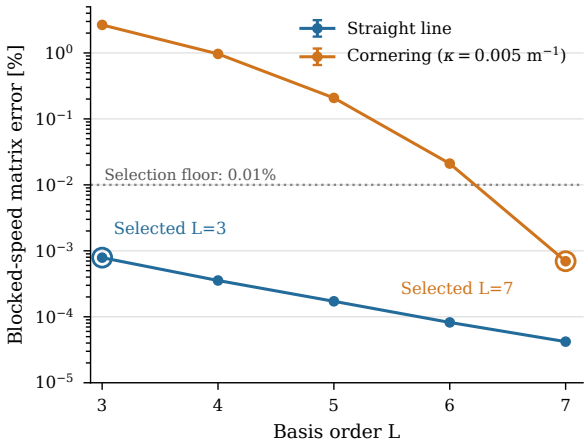
$$F_a = \mu F_{za} \tanh(C_a \alpha_a / (\mu F_{za})), \quad a \in \{f, r\}. \tag{12}$$

Here m is mass, I_z is yaw inertia, and l_f, l_r are the distances from the center of mass to the front/rear axles. For axle a, F_a, F_{za}, C_a, and α_a denote lateral force, normal load, cornering stiffness, and slip angle, respectively, with $\alpha_f = u - \beta - l_f r/v$, $\alpha_r = -\beta + l_r r/v$, $F_{zf} = mgl_r/(l_f + l_r)$, and $F_{zr} = mgl_f/(l_f + l_r)$. Gravity is $g = 9.81$ m/s^2 and the friction coefficient is $\mu = 0.9$. Angles β, u are in radians and r is in rad/s.

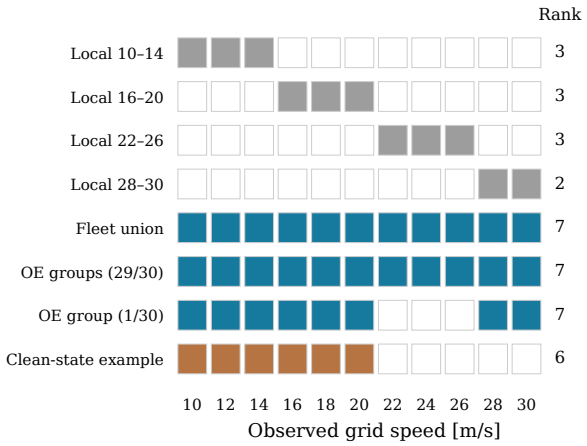
Each fleet contains 30 vehicles, with ten from each parameter center in Table I. Mass and inertia have independent uniform scatter of $\pm 2.5\%$, axle stiffnesses $\pm 5\%$, and geometry is fixed. Over $\mathcal{V} = [10, 30]$ m/s, the local blocks $\{10, 12, 14\}$, $\{16, 18, 20\}$, $\{22, 24, 26\}$, and $\{28, 30\}$ m/s are randomly assigned to eight, eight, seven, and seven clients. Family labels enter neither this allocation nor learned fitting. Each fleet therefore supplies 83 local vehicle–speed datasets.

B. Trajectory acquisition and local OE fitting

Each observed speed supplies a 44-s maneuver starting at $\beta = r = 0$. An LTI model of the Nominal vehicle (Table I), linearized at straight driving and 20 m/s, supplies one LQI gain and prefilter for all vehicles and speeds. The design weights (β, r, ξ) by $Q = \mathrm{diag}(25, 50, 1000)$ and steering by $R_u = 0.15$, where ξ integrates measured yaw-rate error. The integral is limited to ± 0.6 rad with conditional anti-windup; steering is limited to $\pm 12^\circ$ and 30°/s.



(a) Development-only basis validation



(b) Scheduling information ($L = 7$)

Fig. 2. Scheduling representation and information coverage. (a) Earlier blocked-speed matrix error across ten development fleets; error bars denote one standard error. Rings mark the selected orders, and the dotted line is the 0.01% practical selection floor. (b) Filled cells mark observed grid speeds. The 30 primary OE groups have two coverage patterns, both with rank seven. The brown row is a secondary clean-state group in confirmation fleet 5 with rank six. Full rank does not remove within-group mismatch or guarantee extrapolation accuracy.

TABLE I
VEHICLE PARAMETER CENTERS USED TO GENERATE THE FLEET. ALL CLIENTS HAVE $l_f = 1.2$ M AND $l_r = 1.6$ M.

Center	m [kg]	I_z [kg m^2]	C_f	C_r
Nominal	1500	2500	80000	80000
Heavy	1725	3000	84000	80000
Handling	1425	2250	72000	88000

Stiffnesses in N/rad.

The yaw-rate reference ramps to $v\kappa_0$ over the first 4 s, with $\kappa_0 = 0.005$ m^{-1}. After 20 s for settling and a further 4-s hold, a 20-s fitting window tracks

$$r_{\mathrm{ref}}(t) = v\kappa_0 + \frac{a_{\mathrm{exc}}}{v} h(t), \qquad a_{\mathrm{exc}} = 0.3 \text{ m/s}^2, \tag{13}$$

where t is time within the fitting window. The unit-peak waveform h sums sinusoids at 0.2, 0.5, 0.9, and 1.4 Hz with seeded random phases and 2-s sine-squared tapers. Validation uses a separate maneuver with a 0.15–1.2 Hz chirp and the same taper. Feedback generates steering; sideslip is not prescribed independently.

At $T_s = 0.01$ s, each fitting window supplies $N_i(v) = 2000$ consecutive transitions: 2001 measured state vectors and 2000 applied steering inputs. Measurement noise is independent zero-mean Gaussian across samples and components, with standard deviations $0.01°$ for β and $0.03°$/s for r. Feedback uses these noisy states; OE uses the same states and known applied steering, without input measurement noise. Fourth-order Runge–Kutta integrates the plant continuously within each maneuver.

OE uses trust-region least squares with analytic sensitivities and 300 function evaluations per start: affine OLS (initial spectral radius capped at 0.995), $A = 0.9I_2$, $A = 0.98I_2$, and the nominal model. Fitted matrices have no stability constraint. Validation assesses prediction and acquisition feasibility, not start or map selection. The study assumes sideslip sensing and uses simulated trajectories.

C. LPV representation and comparison methods

The common seven-function basis is

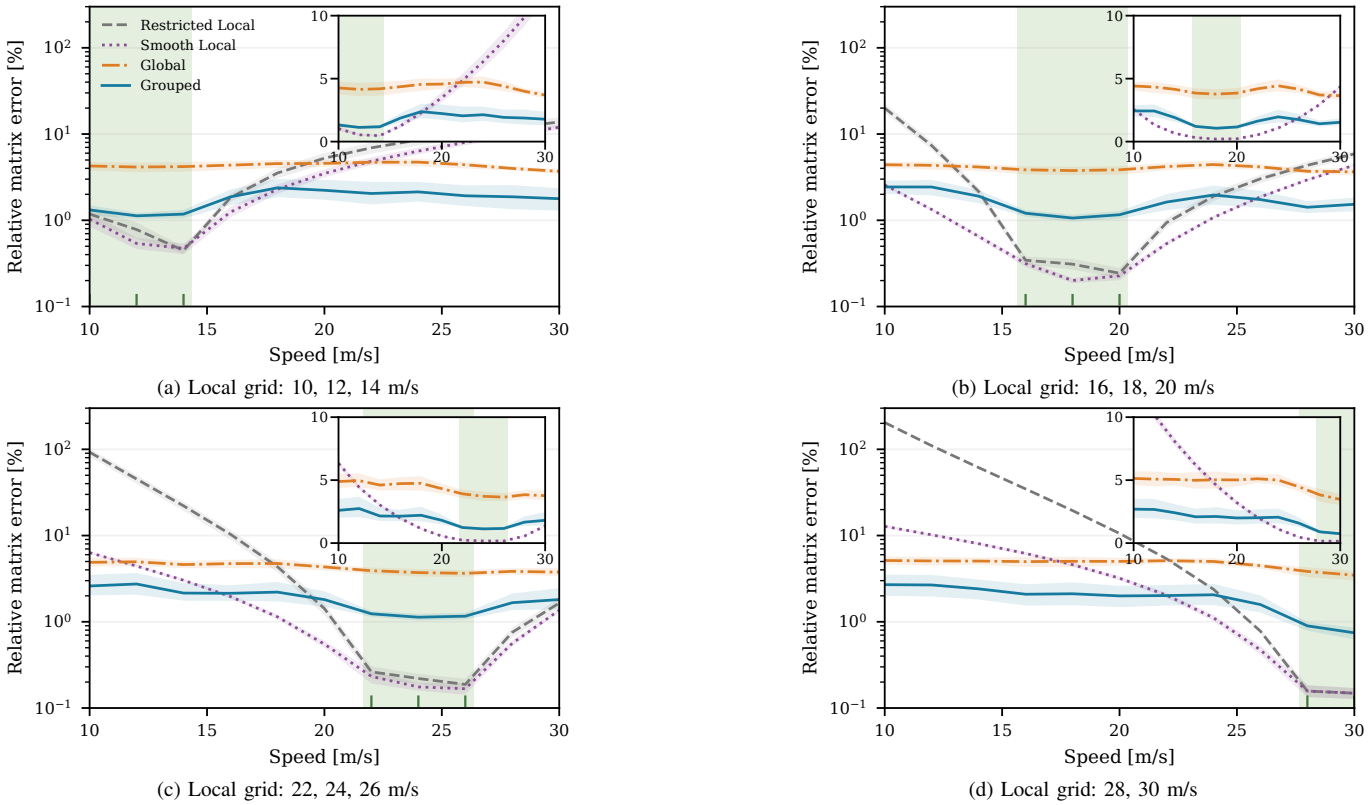
$$b(v) = [1, v^{-1}, v^{-2}, z, z^2, z^3, z^4]^\top, \quad z = (v - 20)/10. \tag{14}$$

Stacking $b(v)^\top$ on 401 uniformly spaced speeds gives $B_{\mathrm{ref}} = QR$, with column-orthonormal $Q \in \mathbb{R}^{401\times 7}$ and invertible $R \in \mathbb{R}^{7\times 7}$. We use $\phi(v)^\top = b(v)^\top R^{-1}$ to improve conditioning without changing the span. Earlier development-only blocked-speed validation of reference matrices selected $L = 7$ for cornering with a one-standard-error rule and error floor 10^{-4}, and $[1, v^{-1}, v^{-2}]$ for straight driving.

Restricted Local fits these three functions to each client's targets with ridge 10^{-8}. Global pools the fleet targets with $L = 7$, while learned groups follow Section III. Independent pooled augmented least squares checks Global and the same learned groups: maximum absolute coefficient and grid-prediction differences must not exceed 10^{-8}. Smooth Local tests whether sharing adds value beyond a different local prior, using $L = 7$ and the curvature-penalized fit

$$\widehat{\Theta}_i^{\mathrm{S}} = \arg\min_{\Theta} \Big\{ \frac{\|Y_i - \Phi_i\Theta\|_F^2}{n_i} + \alpha\|Q_2\Theta\|_F^2 + \eta\|\Theta\|_F^2 \Big\}. \tag{15}$$

Here $Q_2 \in \mathbb{R}^{401\times 7}$ factors the 401-point trapezoidal approximation of $\frac{1}{2}\int_{-1}^{1} \|d^2[\phi(v(z))^\top\Theta]/dz^2\|_2^2\, dz$, where $v(z) = 20 + 10z$, and $\eta = 10^{-16}$ is numerical ridge. Affine functions in z remain unpenalized; the prior does not restore data rank. Earlier development selected α from $\{10^{-6}, 10^{-4}, 10^{-2}, 1, 10^2, 10^4, 10^6\}$ using unobserved-speed matrix error. The largest penalty within one standard error of the best fleet mean was 10^6, approaching an affine speed map. OE confirmation inherits this penalty; all seven strengths form a secondary sensitivity check.



(a) Local grid: 10, 12, 14 m/s

(b) Local grid: 16, 18, 20 m/s

(c) Local grid: 22, 24, 26 m/s

(d) Local grid: 28, 30 m/s

Fig. 3. OE confirmation matrix error by local speed block. Light shading marks each observed block; short green marks identify its observed grid speeds. Curves average clients within block and fleet, then across ten fleets; bands are pointwise 95% fleet-bootstrap intervals from 20,000 draws, used descriptively. Insets show Global, Grouped, and Smooth Local on a linear 0–10% scale with identical averaging and bands. Smooth Local uses the frozen curvature penalty. These bands are distinct from the adjusted paired intervals for the primary endpoint.

D. Evaluation metrics and statistical protocol

Reference matrix vectors $m_i^\star(v)$ come from numerical continuous-time Jacobians at $r_i^\star = v\kappa_0$, $\dot\beta = \dot r = 0$, discretized by an augmented matrix exponential at T_s. True parameters determine these evaluation references, not the acquisition controller or OE fits. On $\mathcal{V}_{\text{grid}} = \{10, 12, \dots, 30\}$ m/s, define the relative matrix error and primary unobserved-speed fleet error as

$$e_i(v) = \frac{\|\widehat{m}_i(v) - m_i^\star(v)\|_2}{\|m_i^\star(v)\|_2}, \qquad E_s^{\mathrm{U}} = \frac{1}{M}\sum_{i=1}^{M} \frac{1}{|\mathcal{U}_i|} \sum_{v\in\mathcal{U}_i} e_i(v), \tag{16}$$

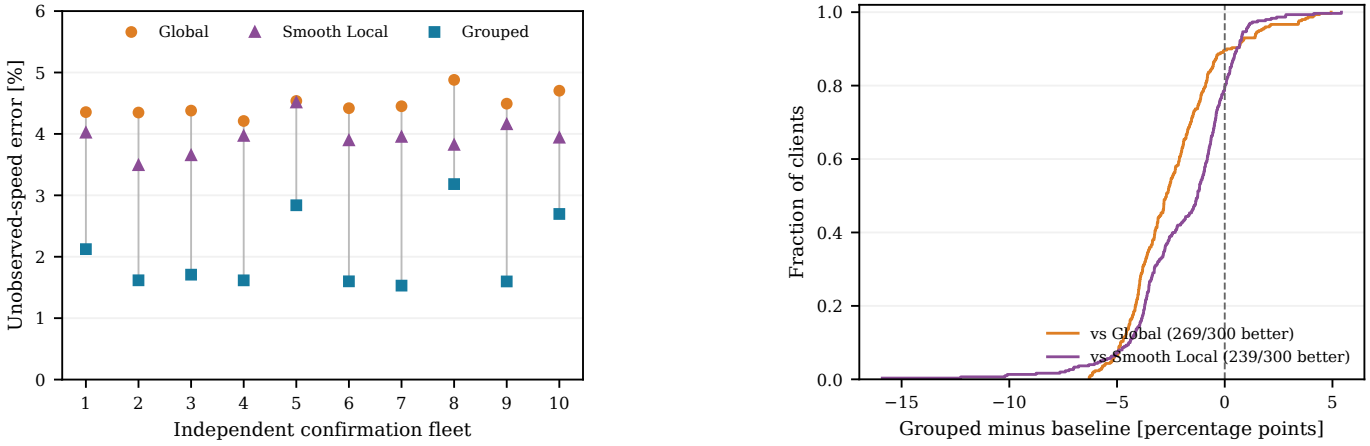
where $\mathcal{U}_i = \mathcal{V}_{\text{grid}} \setminus \mathcal{V}_i$ and s indexes fleets ($M = 30$). Speeds are averaged within clients, clients equally within fleets, and fleets equally across the experiment. Observed-speed and full-grid errors replace $\mathcal{U}_i$ by $\mathcal{V}_i$ and $\mathcal{V}_{\text{grid}}$, respectively. The matrix metric depends on state units; separate relative A and B errors and entrywise absolute errors aid interpretation. None measures tracking quality.

For each primary baseline $b \in \{\text{Smooth Local}, \text{Global}\}$, the paired difference is $\Delta_s^{(b)} = 100(E_s^{\mathrm{U},b} - E_s^{\mathrm{U,Fed}})$ in percentage points. Two Bonferroni-adjusted 97.5% two-sided Student-t intervals give simultaneous 95% family confidence under the paired-t assumptions, with fleets as sampling units. Confirmation requires both lower bounds above zero, ten valid fleet fits, and all equivalence checks passing. Inference with ten fleets is approximate; paired differences and win counts are reported. Percentage gains $100(1 - E_s^{\mathrm{U,Fed}}/E_s^{\mathrm{U},b})$ are averaged over fleets and reported descriptively.

E. Validity checks and secondary diagnostics

Both training and validation maneuvers must satisfy steering limits, $|\beta| \le 5°$, $|a_y| \le 6$ m/s^2, and settling checks during the last 2 s before fitting, where $a_y = v(\dot\beta + r)$ is lateral acceleration. Settling requires mean yaw error at most $\max\{10^{-3}\,\text{rad/s}, 0.01v\kappa_0\}$ and absolute fitted slopes of $\beta/(0.25°)$ and $r/(2°/\text{s})$ at most $0.02\,\text{s}^{-1}$. Recorded actuator limiting is allowed because OE uses the applied input; unsaturated or smooth steering is not required. A failed fit, bound, or settling check blocks that condition for the entire fleet. No cells or fleets are excluded or replaced; fitted maps receive no stability projection or Global fallback.

Secondary fits use OLS targets from noise-free simulator states and applied inputs of the same noisy-feedback trajectories, centered by preceding quiet-window means. Known-family pooling and full-grid reference fits use privileged information. Group coverage, rank, conditioning, and spectral radii diagnose information loss and unstable frozen-speed models. These checks reuse confirmation fleets without changing the primary



(a) Paired primary results

(b) Client tradeoff (descriptive)

Fig. 4. OE confirmation fleet gains and client tradeoffs. (a) Paired unobserved-speed errors; connectors join methods within each independent fleet. (b) Empirical cumulative distributions of client mean unobserved-speed error changes (Grouped minus baseline) in percentage points. Negative values indicate improvement. Client comparisons are descriptive; the 300 clients are nested within ten fleets.

method. This protocol tests identification. Section VI separately evaluates control using the same frozen maps and fleets.

V. Identification Results

A. Local fits and complementary coverage

All 830 local OE fits are valid, and all acquisition runs satisfy the applied-bound and settling checks. Local matrix error has median 0.247% and 95th percentile 1.206%. The steering limiter is active during 6.37% of validation samples on average; these results therefore concern known applied inputs, including limited inputs, rather than unsaturated acquisition.

Accurate local matrices do not resolve missing speed information. The earlier development study selects seven basis functions for cornering, compared with three for straight driving (Fig. 2(a)). The local speed blocks have scheduling ranks three, three, three, and two, whereas their fleet union has rank seven. Figure 2(b) separates this local deficiency from the coverage of learned groups.

B. Independent confirmation of sharing gains

Table II reports mean errors with the equal-client and equal-fleet weighting in Section IV. Grouped fitting reduces unobserved-speed error to 2.05%, compared with 3.94% for Smooth Local and 4.48% for Global. The paired absolute reductions are 1.89 percentage points against Smooth Local, with adjusted interval $[1.39, 2.39]$, and 2.43 points against Global, $[2.02, 2.83]$. Both lower bounds are positive, all ten fleets are valid, and all equivalence checks pass, satisfying the prespecified confirmation criterion. The corresponding mean paired percentage gains are 48.0% and 54.5%.

Local methods fit observed speeds more closely, but grouped sharing improves prediction outside those blocks (Fig. 3). Figure 4(a) shows improvement against both primary baselines in every fleet. Benefits are not uniform across recipients: 269/300 clients improve over Global and 239/300 over Smooth Local, with 31 and 61 worsening, respectively. There are

TABLE II
OE confirmation: mean relative matrix error [%] across ten independent fleets. Unobserved speed is the primary endpoint.

Method	Full grid	Observed	Unobserved
Restricted Local	16.02	0.38	20.44
Smooth Local	3.07	0.33	3.94
Global	4.34	3.87	4.48
Federated groups	1.82	1.09	2.05

no numerical ties at tolerance 10^{-12} in dimensionless error. Figure 4(b) shows the distributions of these changes. Client counts are descriptive because clients share learned maps and are nested within fleets.

The independent pooled solve for the same assignments differs from the federated fit by at most 5.12×10^{-14} in coefficients and 2.24×10^{-14} in grid predictions. This verifies fixed-partition regression equivalence, not agreement between independent grouping searches.

C. What the learned groups capture

Every confirmation fleet selects $K = 3$. All 30 group designs have rank seven, with condition numbers between 3.10 and 16.02 in the fixed QR basis. Twenty-nine groups contain all eleven grid speeds. The remaining group lacks observations at 22–26 m/s but has observations on both sides; its mean unobserved-speed error is 1.32%. Thus full scheduling rank does not require observing every grid point. These are properties of the selected designs, not guarantees supplied by minimum group size or BIC.

Family labels, inspected only afterward, give adjusted Rand indices from 0.518 to 1. Exact family recovery occurs in 4 of ten fleets. The other partitions mix physical families despite selecting three groups. The benefit therefore concerns sharing among vehicles selected from their observed dynamics, rather than reliable recovery of the generating vehicle classes.

TABLE III
PRIMARY OE DIAGNOSTICS. A AND B COLUMNS ARE MEAN RELATIVE ERRORS [%] AT UNOBSERVED SPEEDS. THE LAST COLUMN COUNTS MATRICES WITH SPECTRAL RADIUS AT LEAST ONE OVER ALL 3300 CLIENT–GRID PAIRS.

Method	A error	B error	Unstable
Restricted Local	18.72	28.23	136
Smooth Local	2.73	10.98	272
Global	4.13	6.50	0
Federated groups	1.75	3.68	0

D. Sensitivity and limits of the result

Across the seven prespecified Smooth Local penalties, grouped fitting improves every fleet–penalty comparison (70/70). The best aggregate confirmation penalty, 10^{-2}, gives 3.88% unseen error, compared with 3.94% for the frozen 10^6 penalty. The mean paired gain against this retrospectively best setting remains 47.3%. This sensitivity check does not replace the frozen baseline or constitute another independent confirmation.

Table III shows lower errors for both A and B; the improvement is not confined to the combined matrix metric. Aggregate absolute error also falls for each of the six entries against both primary baselines. All grouped frozen-speed matrices have spectral radius below one, with maximum 0.9765. This check neither certifies stability under varying speed nor establishes closed-loop performance.

Clean-state OLS reduces median local matrix error to 0.052%, but more accurate local targets do not ensure a better group partition. In one fleet, a selected five-client group covers only 10–20 m/s and has rank six (Fig. 2(b)). Its unobserved-speed error is 38.76%, raising the fleet mean to 9.68%. No fallback is applied. Changing the local targets also changes the assignments, so this outcome does not establish that measurement noise helps identification.

Known-family pooling is likewise not an accuracy bound. Two fleets have mean unseen errors of 44.19% and 23.35% when a physical family lacks observations at 10–14 m/s. These family designs remain full rank but have condition numbers above 3000. Together, these diagnostics distinguish accurate local estimation, sufficient group coverage, and reliable extrapolation. They expose failure modes without establishing robustness to other noise levels, coverage distributions, or real driving data.

VI. SUPPORTING LQI VALIDATION

We next test whether the OE-derived maps support useful nonlinear tracking under a common controller design. All models and assignments from the ten confirmation fleets in Section V remain fixed. This is a supporting control comparison on those same 300 vehicles, not a new independent confirmation. No identification, grouping, or controller-weight selection is repeated.

A. Common controller design

The comparison includes Restricted Local, Smooth Local, Global, and federated group maps. Nominal LTI uses exactly the fixed 20-m/s gain and prefilter that generated the identification trajectories in Section IV-B. ExactLPV is a privileged recipient-specific $L = 7$ fit to the full reference matrix grid, not a guaranteed performance bound. All scheduled methods evaluate their saved maps at 10, 12, …, 30 m/s. In particular, Restricted Local extrapolates its three-function map beyond its observed block without a nearest-speed fallback.

At each grid node, the fitted $\widehat{A}, \widehat{B}$ define

$$A_{\mathrm{a}} = \begin{bmatrix} \widehat{A} & 0 \\ T_s c_r & 1 \end{bmatrix}, \quad B_{\mathrm{a}} = \begin{bmatrix} \widehat{B} \\ 0 \end{bmatrix}, \quad c_r = [0, 1]. \tag{17}$$

The augmented state is $[\beta, r, \xi]^\top$, where ξ integrates yaw-rate error. Discrete Riccati synthesis uses the acquisition weights $Q = \mathrm{diag}(25, 50, 1000)$ and $R_u = 0.15$ for every method. For unit yaw reference, the equilibrium state $h \in \mathbb{R}^2$ and input $w \in \mathbb{R}$ solve $(I - \widehat{A})h - \widehat{B}w = 0$ and $c_r h = 1$. The prefilter is $F = w + K_x h$, where K_x contains the first two gain entries. Gains K and prefilters F are linearly interpolated between grid nodes; Nominal LTI remains constant across speed.

For yaw reference r_k^{ref}, the runtime law is

$$\begin{aligned} u_k^{\circ} &= -K(v_k)[\beta_k, r_k, \xi_k]^\top + F(v_k) r_k^{\mathrm{ref}}, \\ \xi_{k+1}^{\mathrm{cand}} &= \mathrm{clip}_{[-0.6, 0.6]}\{\xi_k + T_s(r_k - r_k^{\mathrm{ref}})\}. \end{aligned} \tag{18}$$

Steering is first rate-limited relative to u_{k-1}, then amplitude-limited. The integral candidate is accepted when the command is unsaturated or the increment counteracts limiting; otherwise $\xi_{k+1} = \xi_k$. The integral bound is in radians. Feedback uses noise-free physical states, isolating the effect of models learned from noisy identification data. Homogeneous A, B maps supply feedback and the prefilter; fitted OE offsets and initial states are not used online.

B. Tracking protocol

Each nonlinear maneuver lasts 12 s and starts at $x_0 = 0$, $\xi_0 = 0$, and $u_{-1} = 0$. RK4 integration holds applied steering and prescribed speed fixed within each $T_s = 0.01$ s step. The moderate maneuver varies speed smoothly as $11 \to 29 \to 15$ m/s and uses an enveloped two-frequency lateral-acceleration demand with scale 4 m/s^2 and frequencies 0.35 and 0.72 Hz. The hard maneuver varies speed as $29 \to 12 \to 28$ m/s and uses Gaussian demands centered at $3.0, 6.0, 9.2$ s, with widths $0.40, 0.48, 0.55$ s and amplitudes $7, -7.7, 5.6$ m/s^2, respectively. Yaw reference is requested acceleration divided by speed.

All methods share steering limits of 12° and 120°/s and a 6° sideslip bound. The control test thus permits a higher steering rate than the 30°/s acquisition limit. Limiter activation is recorded separately from violations of applied-input or sideslip bounds. Finite infeasible responses remain in the complete-maneuver yaw RMSE, averaged first over clients and then fleets. The changing turn demand tests usability beyond the identification cornering condition, without validating full path tracking or observer-based control.

TABLE IV

OE-BASED SUPPORTING CONTROL: MEAN YAW-RATE RMSE [DEGREES/S] OVER TEN REUSED FLEETS. VIOLATIONS COUNT CLIENTS EXCEEDING 6° SIDESLIP (MODERATE/HARD), OUT OF 300 EACH. ALL FINITE RESPONSES, INCLUDING INFEASIBLE ONES, CONTRIBUTE TO THE MEANS.

Method	Moderate	Hard	Violations
Nominal LTI	0.0540	0.2026	0/0
Restricted Local	6.2386	10.2444	51/77
Smooth Local	4.8596	3.1910	70/62
Global	0.0447	0.1533	0/0
Federated groups	0.0409	0.1513	0/0
ExactLPV reference	0.0400	0.1512	0/0

C. Tracking results and recipient tradeoffs

All 1800 schedules synthesize and all 3600 rollouts finish numerically. Table IV reports tracking errors together with sideslip violations. Nominal LTI, Global, Federated, and ExactLPV satisfy the tested constraints in every rollout. Restricted Local and Smooth Local have substantial tracking tails, including infeasible responses. Extreme sideslip in those responses lies outside credible vehicle operation. Their large means therefore describe failures of the tested control design, not realistic driving performance; satisfying the bounds alone also does not ensure small tracking error.

Mean paired fleet-relative RMSE reductions against Nominal LTI are 24.25% and 25.33% for the moderate and hard maneuvers, with improvements in all ten fleets. Against Global, the reductions are 8.36% and 1.29%, with gains in 10/10 and 8/10 fleets. The corresponding absolute gains are 0.00374 and 0.00203 degrees/s. Percent reductions are averaged across paired fleets and reported descriptively. Steering effort is similar to Global; the hard maneuver uses slightly more steering-rate RMS and limiter activity.

Aggregate improvement is not uniform across recipients. All 100 heavy-family clients improve over Global in both maneuvers, while all 100 handling-family clients worsen. Thus, lower aggregate identification error does not imply better tracking for every vehicle. The maximum federated frozen-point closed-loop radius is 0.9756, evaluated on 81 speeds with the saved ExactLPV map and interpolated gains. Restricted Local and Smooth Local have radii at least one for 157/300 and 77/300 schedules, respectively, despite successful own-model synthesis. These diagnostics concern unsaturated frozen-speed approximations and provide no nonlinear or varying-speed stability guarantee.

VII. DISCUSSION AND CONCLUSION

Federated group regression combines incomplete local speed coverage while limiting the bias of global pooling. With trajectory-based local OE fits, the independent identification confirmation supports lower aggregate unobserved-speed error than both frozen baselines. Benefits do not extend to every recipient, and learned groups need not recover physical families. Agreement with a pooled solve for fixed assignments verifies the distributed regression, not the grouping search or its optimality.

Accurate local estimates, group coverage, and reliable extrapolation are distinct requirements. The secondary clean-state and known-family fits show that improved targets or physical grouping alone can still leave poorly conditioned or deficient group designs. Full rank of the primary selected groups is empirical; neither minimum group size nor BIC guarantees it. Ordinary driving data, observer errors, new-vehicle transfer, and real fleets remain unvalidated.

The frozen OE-derived maps also support feasible scheduled control, with consistent fleet-level tracking gains over the fixed nominal controller. The additional gains over Global are modest and recipient dependent. Both local priors produce failed control responses for some vehicles under the common design. Together, the identification and control results support compatible sharing when pooled coverage is informative, while requiring separate checks of recipient accuracy and closed-loop behavior.

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